

Today's Outline

- 3D Computer Vision

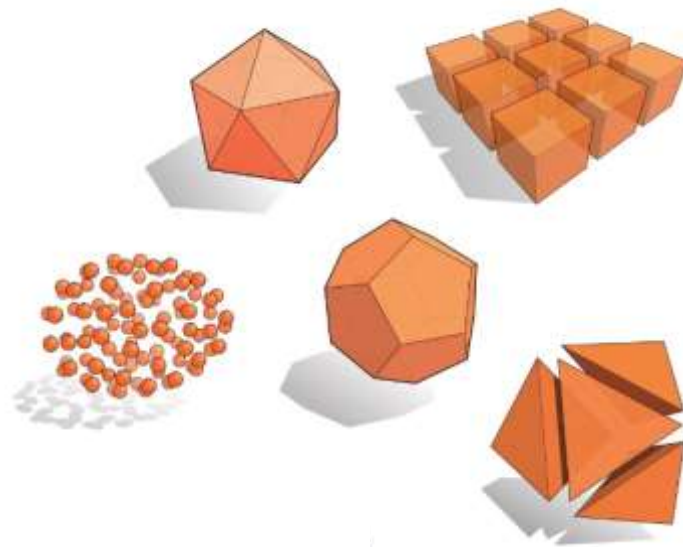
Viewing 3D information from 2D image plane



RGB Image.

Many Ways to Represent Geometry

- Explicit
 - Point cloud
 - Polygon mesh
 - Subdivision
 - ...
- Implicit
 - Level sets
 - Distance functions
 - ...



Each choice best suited to a different task/type of geometry

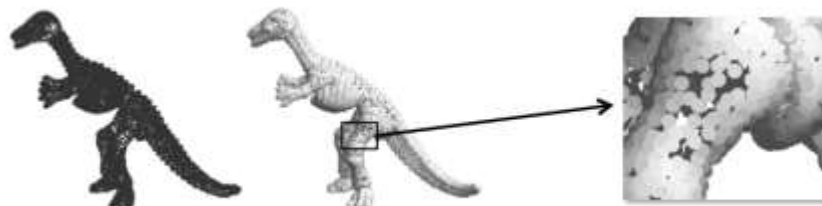
Representation Considerations

- Needs to be stored in the computer
- Creation of new shapes
 - Input metaphors, interfaces...Distance functions
- Operations
 - Editing, simplification, smoothing, filtering, repairing...
- Rendering
 - Rasterization, ray tracing, neural rendering...
- Animation

Point Cloud

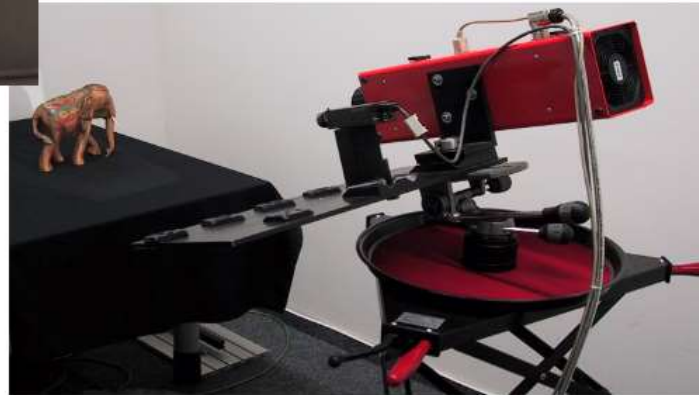
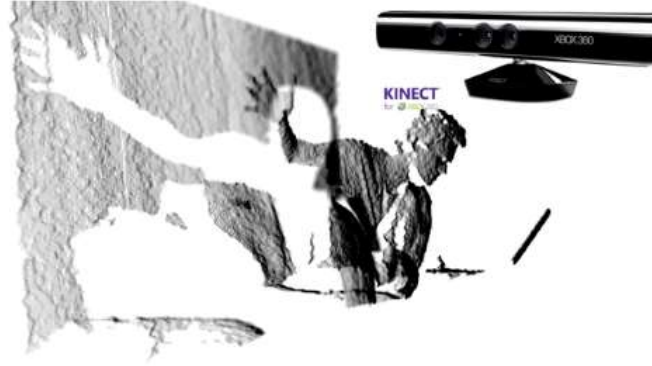
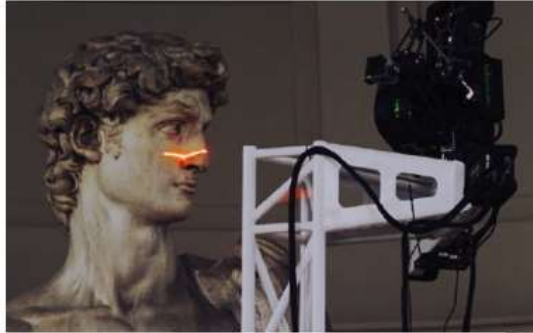


- Simplest representation: only points, no connectivity
- Collection of (x,y,z) coordinates, possibly with normal
- Points with orientation are called surfels



Shading needs normals!

Output of Acquisition



Point Cloud



- Simplest representation: only points, no connectivity
 - Collection of (x,y,z) coordinates, possibly with normal
 - Points with orientation are called surfels
-
- Often results from scanners
 - Potentially noisy
 - Registration of multiple images

Point Cloud



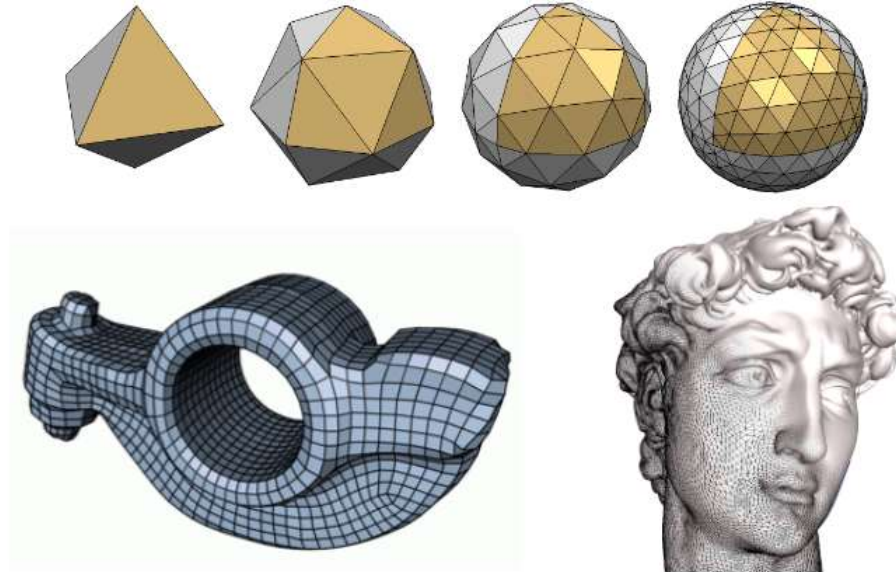
- Simplest representation: only points, no connectivity
- Collection of (x,y,z) coordinates, possibly with normal
- Points with orientation are called surfels

- Other limitations:
 - No simplification or subdivision
 - No direction smooth rendering
 - No topological information



Polygonal Meshes

- Boundary representations of objects



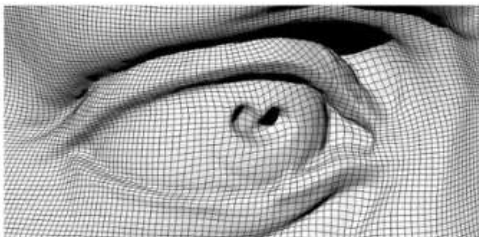
A Large Triangle Mesh

David

Digital Michelangelo Project

28,184,526 vertices

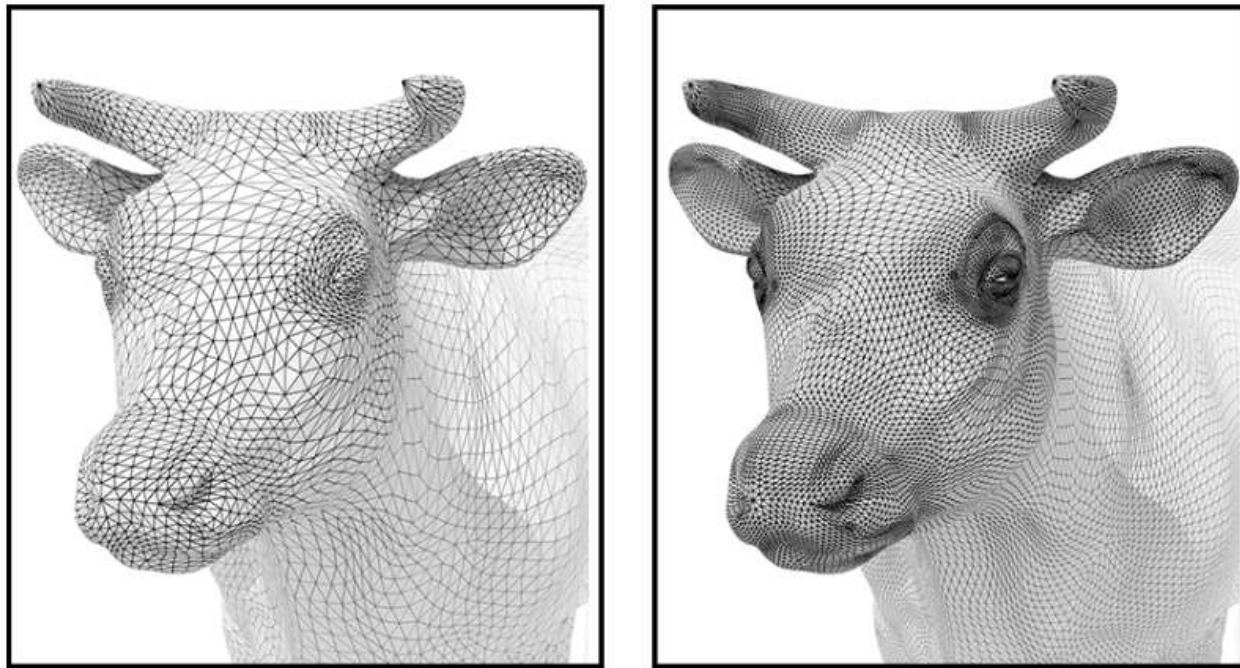
56,230,343 triangles



A Very Large Triangle Mesh

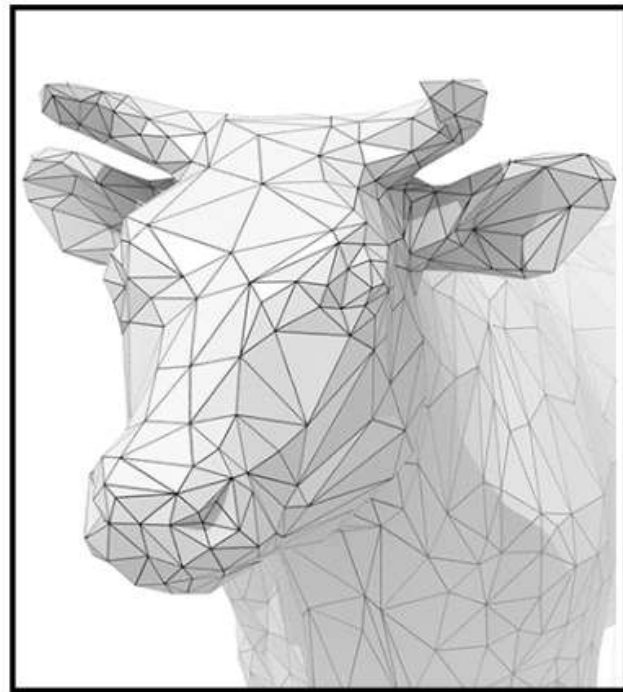
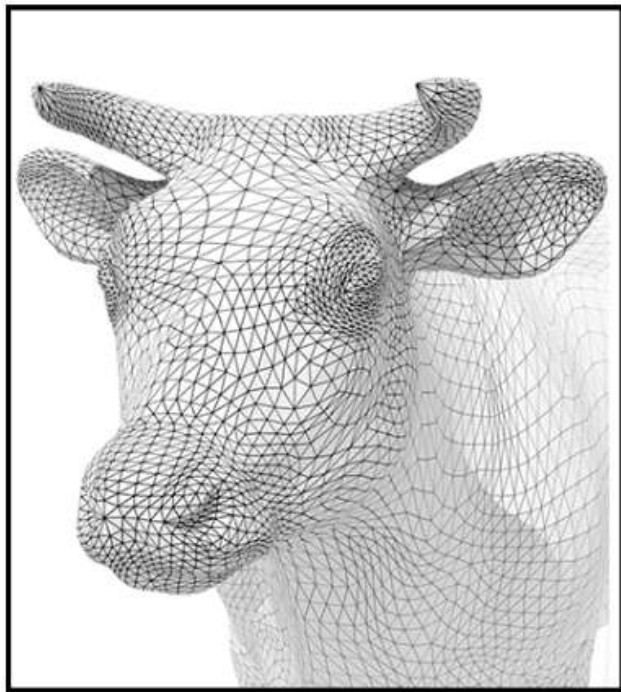


Mesh Upsampling - Subdivision



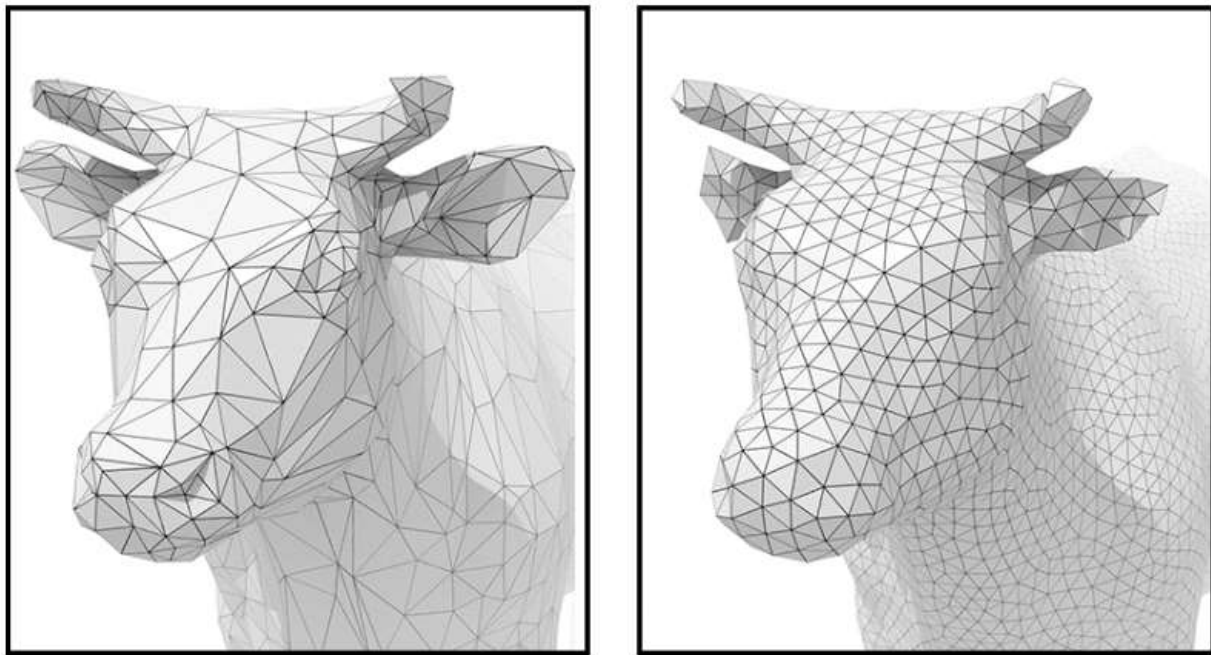
Increase resolution via interpolation

Mesh Downsampling - Simplification



Decrease resolution; try to preserve shape/appearance

Mesh Regularization



Modify sample distribution to improve quality

Shape Representations

Non-parametric



Points

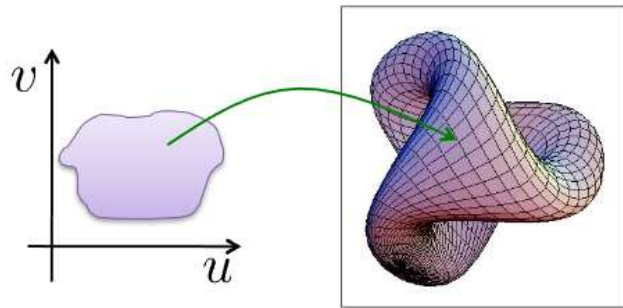


Meshes

Parametric Representations

Range of a function $f : X \rightarrow Y, X \subseteq \mathbb{R}^m, Y \subseteq \mathbb{R}^n$

Surface in 3D: $m = 2, n = 3$



$$s(u, v) = (x(u, v), y(u, v), z(u, v))$$

Parametric Curves

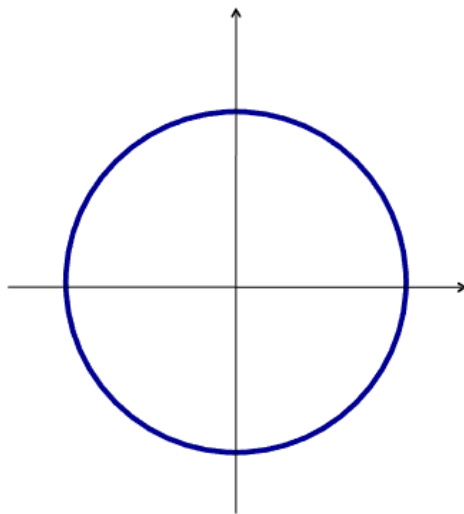
Explicit curve/circle in 2D

$$\mathbf{p} : \mathbb{R} \rightarrow \mathbb{R}^2$$

$$t \mapsto \mathbf{p}(t) = (x(t), y(t))$$

$$\mathbf{p}(t) = r (\cos(t), \sin(t))$$

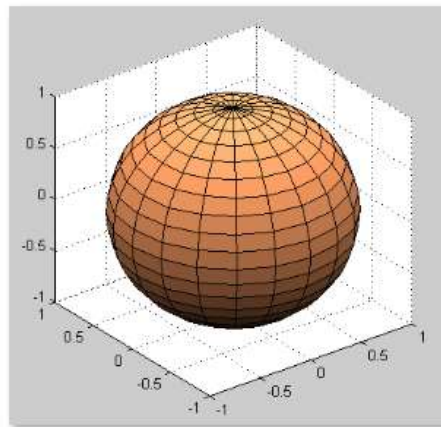
$$t \in [0, 2\pi)$$



Parametric Surface

Sphere in 3D

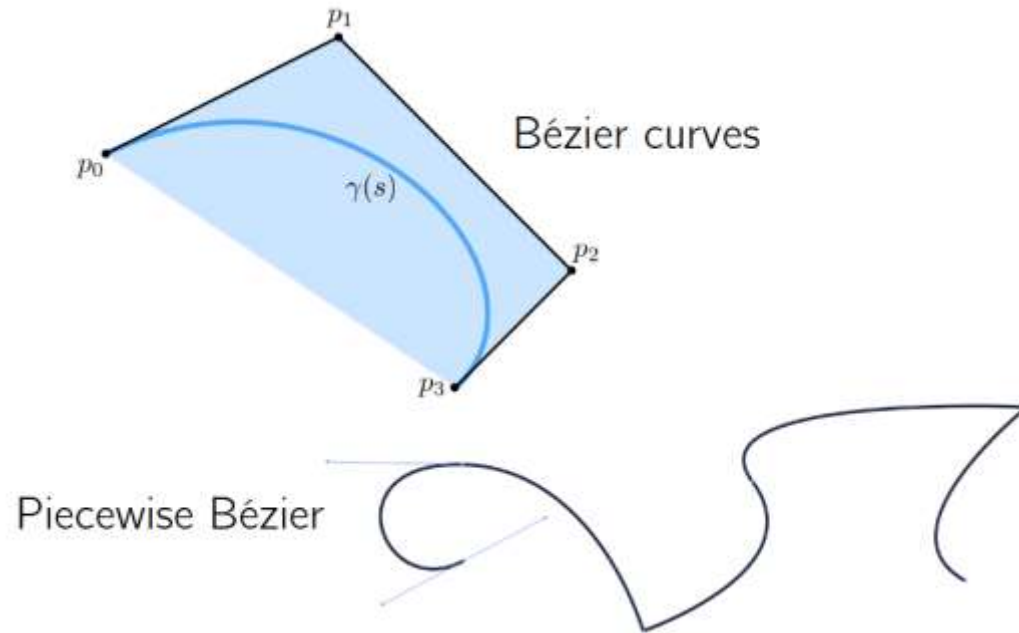
$$s : \mathbb{R}^2 \rightarrow \mathbb{R}^3$$



$$s(u, v) = r (\cos(u) \cos(v), \sin(u) \cos(v), \sin(v))$$

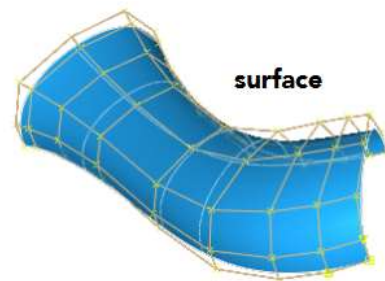
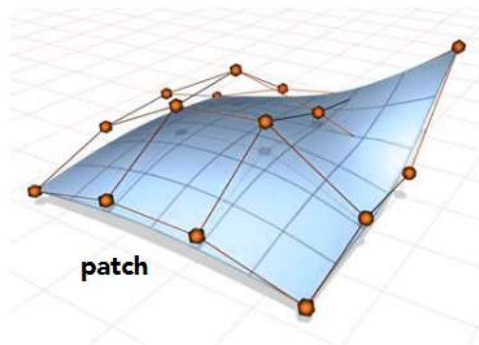
$$(u, v) \in [0, 2\pi) \times [-\pi/2, \pi/2]$$

Bézier Curves



Bézier Surfaces

Use tensor product of Bézier curves to get a patch:



Multiple Bézier patches form a surface.

Shape Representations

Non-parametric

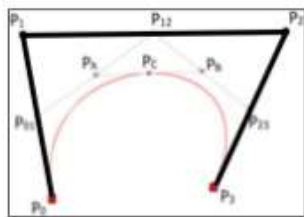


Points

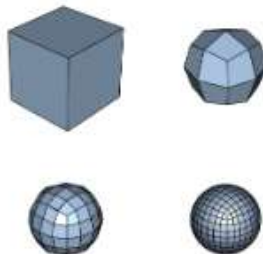


Meshes

Parametric



Splines



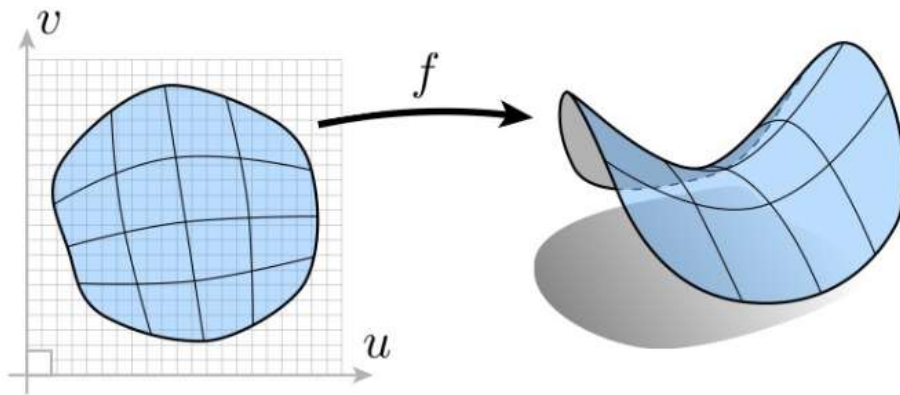
Subdivision
Surfaces

“Explicit” Representations of Geometry

All points are given directly.

Generally:

$$f : \mathbb{R}^2 \rightarrow \mathbb{R}^3; (u, v) \mapsto (x, y, z)$$

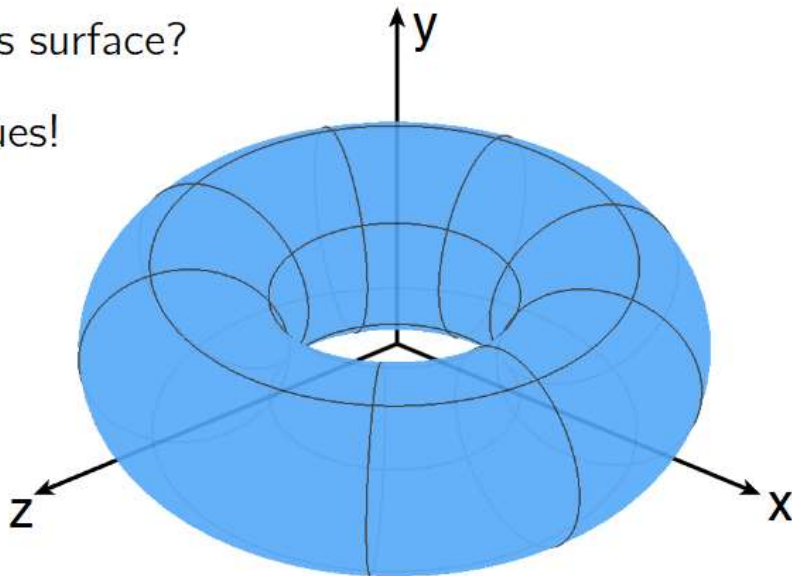


Explicit Surface – Sampling Is Easy

$$f(u, v) = ((2 + \cos u) \cos v, (2 + \cos u) \sin v, \sin u)$$

What points lie on this surface?

Just plug in (u, v) values!



Explicit representations make some tasks easy.

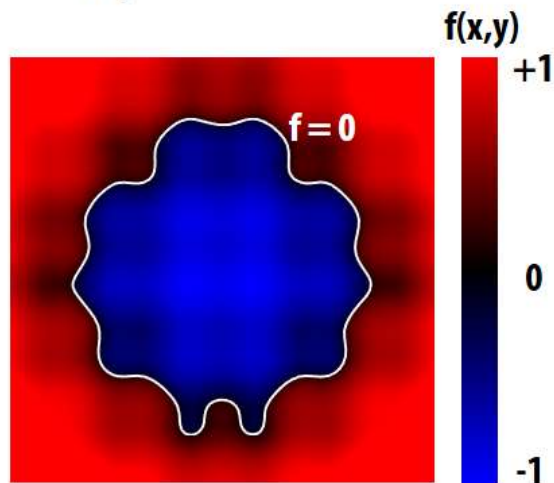
“Implicit” Representations of Geometry

Based on classifying points

- Points satisfy some specified relationship.

E.g., sphere: all points in 3D, where $x^2 + y^2 + z^2 = 1$

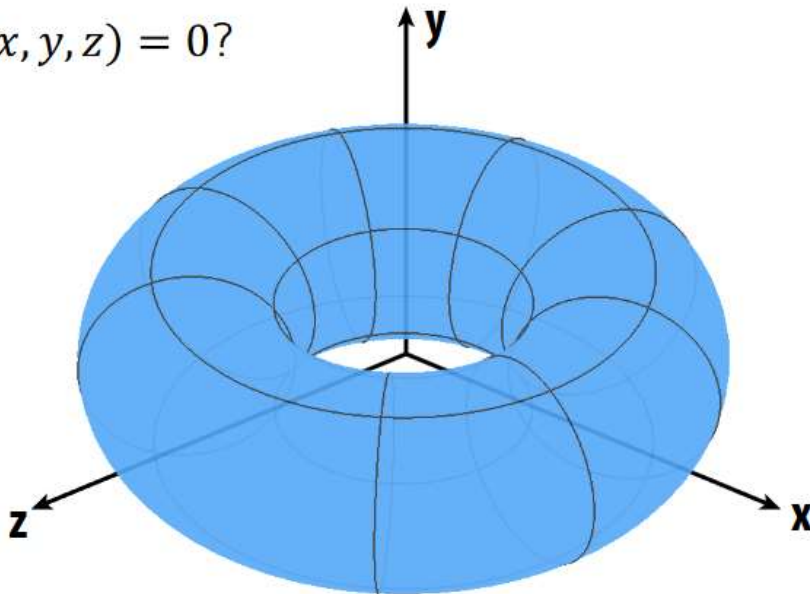
More generally, $f(x, y, z) = 0$



Implicit Surface – Sampling Can Be Hard

$$f(x, y, z) = (2 - \sqrt{x^2 + y^2})^2 + z^2 - 1$$

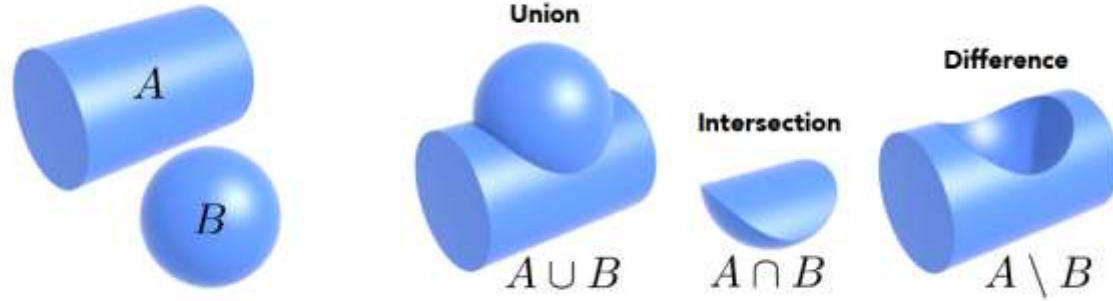
What points lie on $f(x, y, z) = 0$?



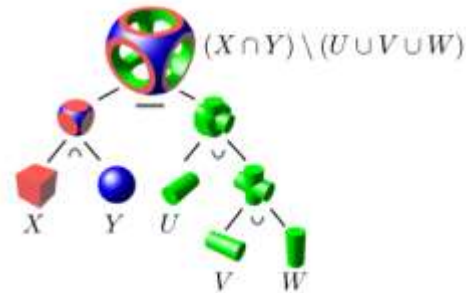
Some tasks are hard with implicit representations.

Constructive Solid Geometry (Implicit)

Combine implicit geometry via Boolean operations



Boolean expressions:

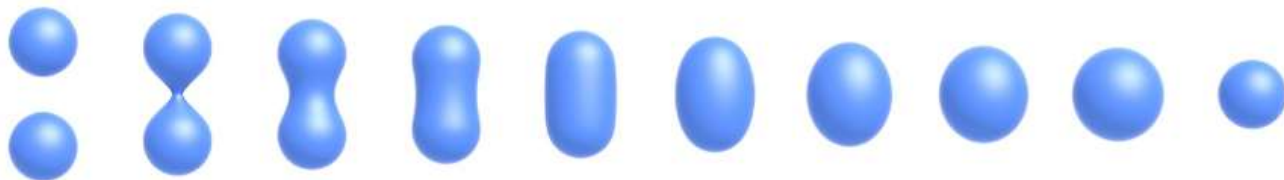


Distance Functions (Implicit)

Instead of Boolean, gradually blend surfaces together using

Distance functions:

Giving minimum distance (could be **signed** distance) from anywhere to object



Shape Representations

Explicit

Implicit

Non-parametric

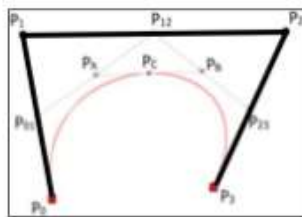


Points

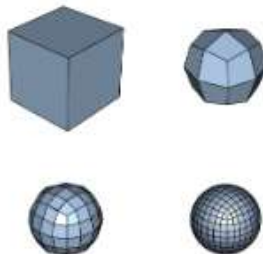


Meshes

Parametric



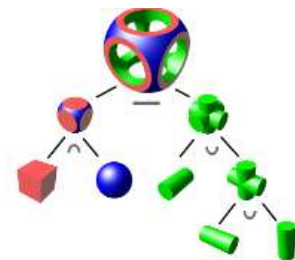
Splines



Subdivision
Surfaces



Algebraic
Surfaces



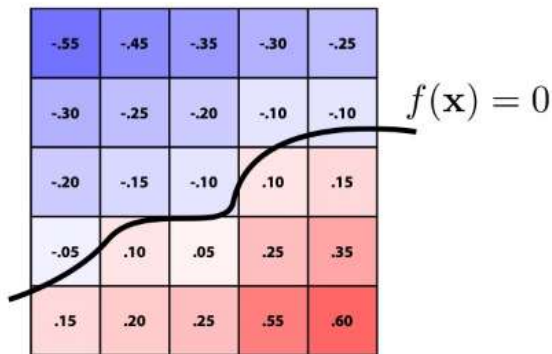
Constructive Solid
Geometry

Level Set Methods (Implicit)

Implicit surfaces have some nice features (e.g., merging/splitting).

But hard to describe complex shapes in closed form

Alternative: store a grid of values approximating function

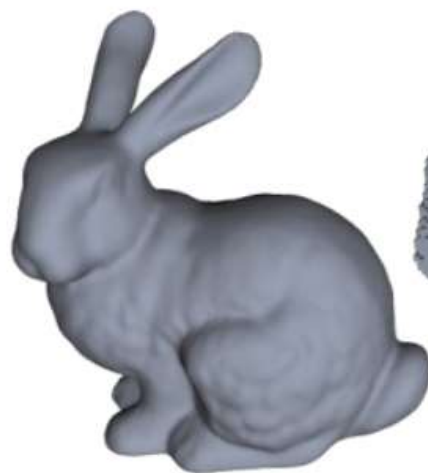
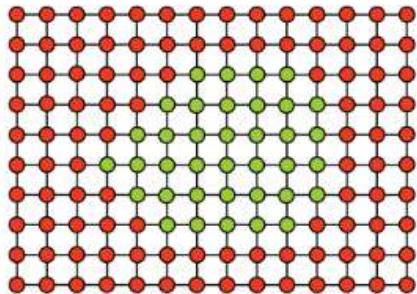
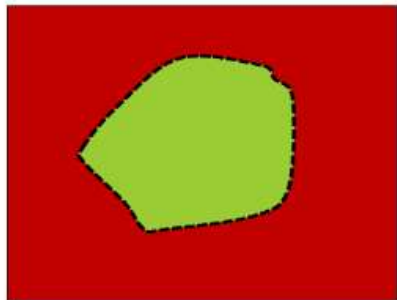


Surface is found where interpolated values equal zero.

Provides much more explicit control over shape (like a texture)

Related Representation: Voxels

- Binary thresholding the volumetric grid

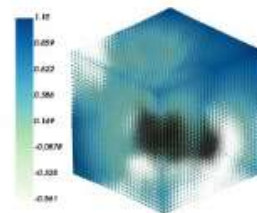


Shape Representations

Explicit

Implicit

Non-parametric



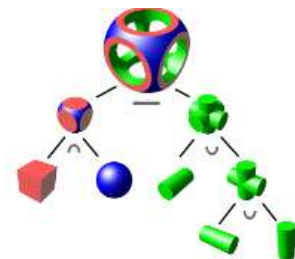
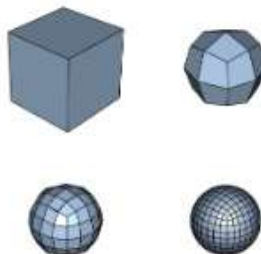
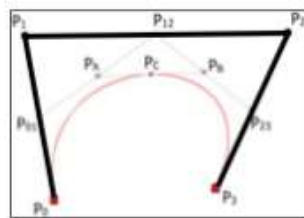
Points

Meshes

Voxels

Level Sets

Parametric



Splines

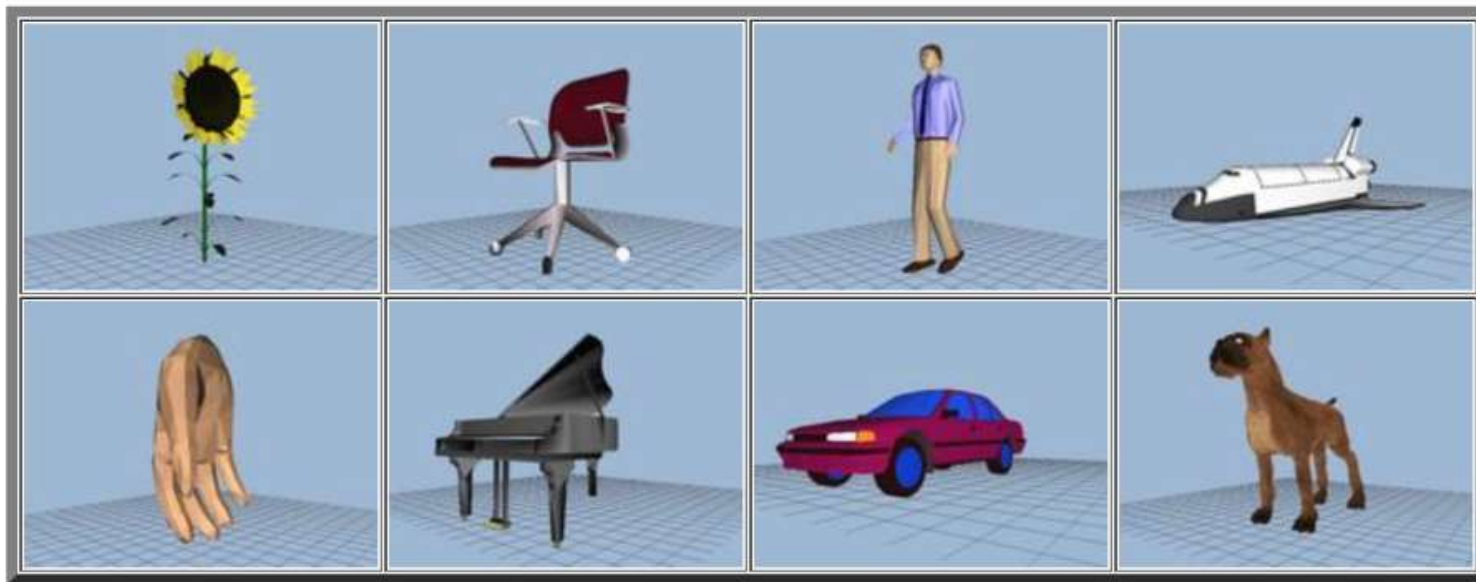
Subdivision
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Constructive Solid
Geometry

Princeton Shape Benchmark

- 1814 Models
- 182 Categories



Datasets for 3D Objects

- Large-scale Synthetic Objects: ShapeNet, 3M models
- ModelNet: absorbed by ShapeNet
- ShapeNetCore: 51.3K models in 55 categories



Objaverse (800K) and Objaverse-XL (10M)



CO3D

- 19,000 videos
- 50 categories



Datasets for Indoor 3D Scenes

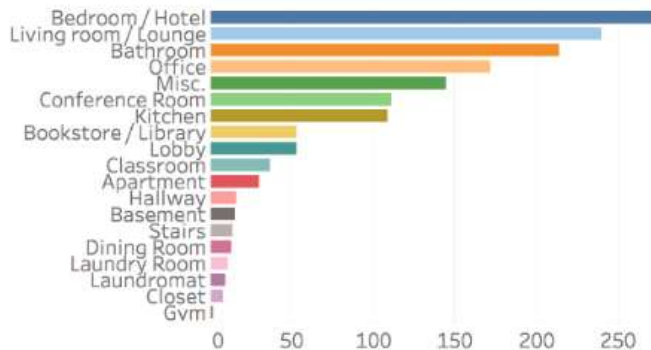
Scanned Real Scenes: ScanNet

- 2.5M Views in 1,500 RGBD scans
- 3D camera poses
- Surface reconstructions
- Instance-level semantic segmentations



Most recently

- ARKitScenes,
- ScanNet++ (with DSLR images)

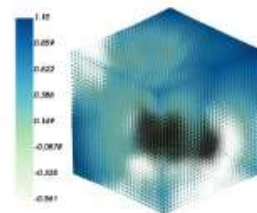


AI + Geometry: Which Representation?

Explicit

Implicit

Non-parametric



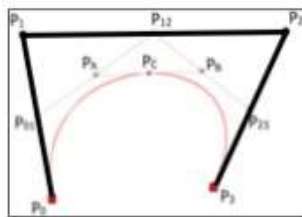
Points

Meshes

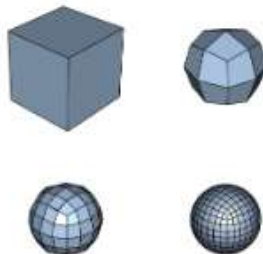
Voxels

Level Sets

Parametric



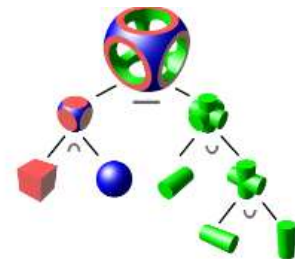
Splines



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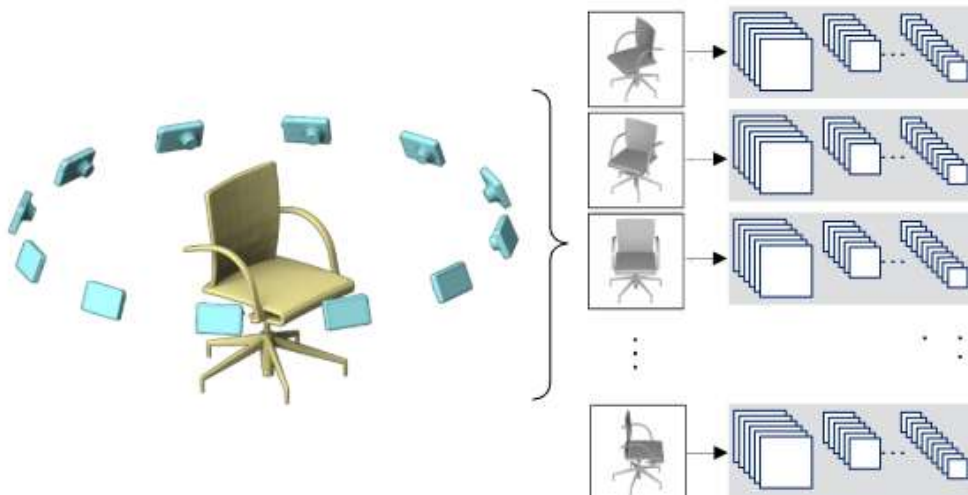


Constructive Solid
Geometry

Multi-View CNN

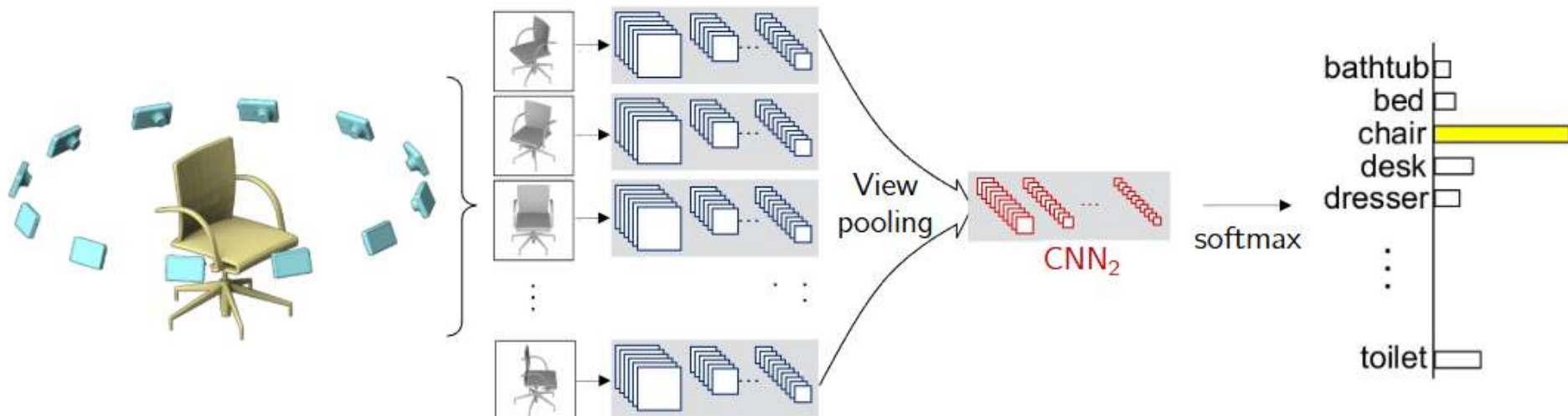


Multi-View CNN



CNN1: feature extraction

Multi-View CNN



CNN_2 : a second ConvNet
producing shape descriptors

Multi-View CNN

Method		Classification (Accuracy)	Retrieval (mAP)
Non-deep {	SPH	68.2%	33.3%
	LFD	75.5%	40.9%
	3D ShapeNets	77.3%	49.2%
	FV, 12 views	84.8%	43.9%
	CNN, 12 views	88.6%	62.8%
	MVCNN, 12 views	89.9%	70.1%
	MVCNN+metric, 12 views	89.5%	80.2%
	MVCNN, 80 views	90.1%	70.4%
	MVCNN+metric, 80 views	90.1%	79.5%

On ModelNet 40

In Reality: Images with Unknown Poses



In Reality: Estimating Pose First



Feature Matching

Match Features Between Each Pair of Images

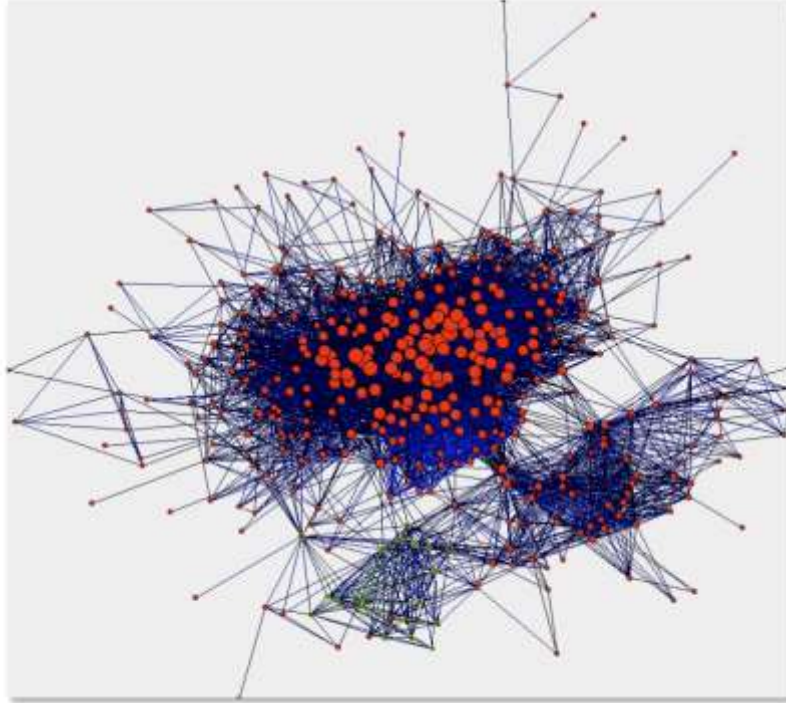


Feature Matching

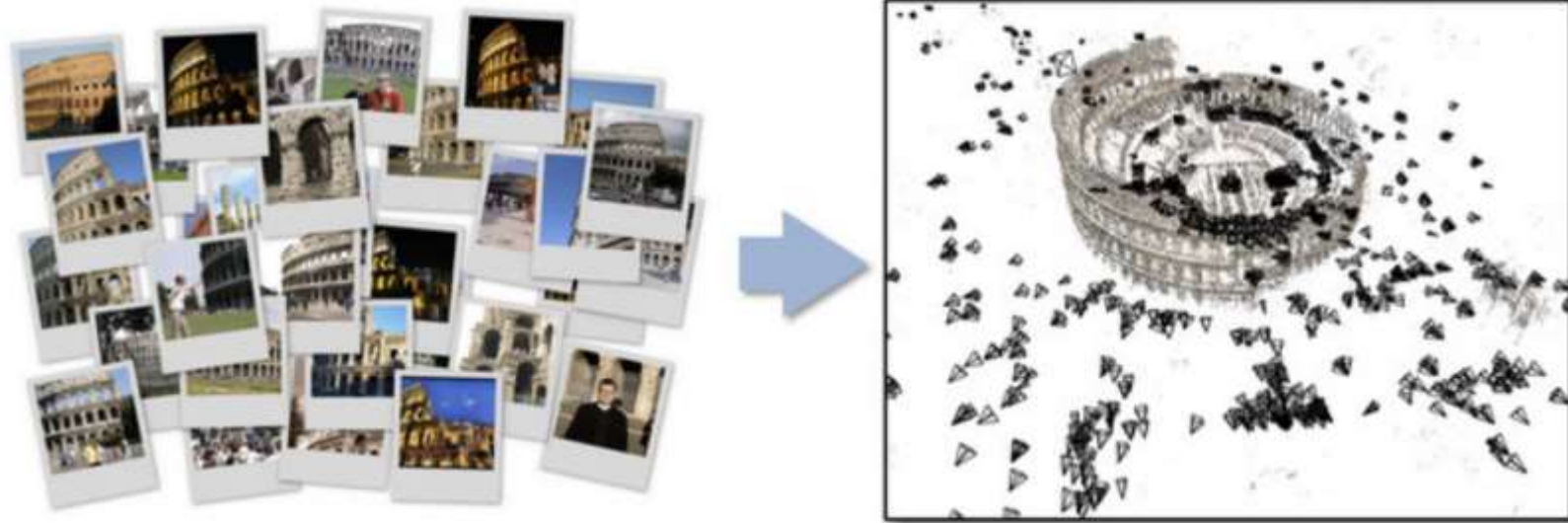
Use RANSAC to estimate fundamental matrix between each pair



Building the Image Connectivity Graph



A Representative 3D Recovery Pipeline



The Full 3D Perception and Reconstruction Pipeline

